



UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment 2 CLO Mapping: CLO-1

Course Code	MAT233	Semester	Fall 2025
Course Title	Linear Algebra		
Instructor	Dr. Muhammad Rayees Ahmad		
Assignment Given Date	03-11-2025	Submission Date	17-11-2025
Total Marks	35 = Assignment (30) + Quiz Based Viva (5)		

Sr.	Name of student	Registration no.	Section	Assignment marks	Viva marks	Total marks
1						

Submission Instructions (Please follow strictly)

- Take the PRINTOUT OF THIS ASSIGNMENT PDF file and solve the question at the given empty space in the assignment PDF file. DO NOT use any other papers.
- This is an INDIVIDUAL assignment.
- Any question related to assignment topic may be asked in the viva.
- Write your complete names and registration numbers on each page of the assignment.
- Follow the deadline. Finish your work one day before, so you could submit in time.
- LATE SUBMISSION WILL NOT BE ACCEPTED (Deduction of 5% marks each day).
- All Plagiarized assignments will be awarded ZERO marks.

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student signature: _____

Name:

Registration No.

Question No.1

[10 marks]

Let $(y + 4)^2 = \frac{1}{2}(x - 2)$ be a parabola. Find the equation of Parabola after Rotating it by an angle of 60° anticlockwise.

Solution:

Question No.2**[CLO1]****[10 marks]**

As a designer for a mobile app, you're tasked with prototyping a simple login screen layout using geometric shapes. Each component (like buttons, input fields, and icons) will be represented using basic shapes such as rectangles and circles.

Design Your UI Layout Using Basic Geometric Shapes

Use the coordinate plane to create a basic mobile login screen with the following components:

- **Screen Container (rectangle):** Vertices at $(-1,1)$, $(-1,8)$, $(3,8)$, $(3,1)$
- **Login Button (triangle):** Vertices at $(1,2)$, $(1.5,3)$, $(2,2)$
- **Username Field (rectangle):** Vertices at $(2,5)$, $(2,6)$, $(0,6)$, $(0,5)$

a) Translation: Move the layout 2 units right and 1 units down to simulate positioning it on a different screen size or resolution.

b) Reflection: Move the layout about the line $y = -x$.

Solution:

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Name:

Registration No.

Name:

Registration No.

Question No. 3**[10 marks]**

You are an undercover cryptographer for a company. Your mission is to encrypt a classified phrase “**Pick the Thing**” using Hill Cipher before transmission to headquarters.

Your Task:

- i. Remove spaces and convert to uppercase, if needed, **pad** the message with an extra letter.
- ii. Convert letters to numbers using the standard alphabet mapping
$$A = 1, \quad B = 2, \quad \dots, \quad Z = 26$$
- iii. Use the following **key matrix** K for encryption
$$K = \begin{bmatrix} 1 & 2 & 9 \\ 2 & 1 & 3 \\ 1 & 2 & 4 \end{bmatrix}$$
- iv. **Encrypt** the message in blocks of **3** letters using **matrix multiplication** modulo **26**.
- v. Convert the **result back to letters** and **write the final encrypted message**.

Solution:

Name:

Registration No.

Name:

Registration No.

Name:

Registration No.

Name:

Registration No.
